Answers to question 1:

1.a Bookwork
Must Explain
-Bandgap Voltage reference Circuit [2]
-PTAT (proportional to absolute temperature) current generator [2]

b.) Figure 2.a of question is a bandgap voltage reference circuit which has almost zero temperature coefficient, used mainly as stable voltage reference in ICs. For BG reference,

$$-V_{BE1} = V_{BE2} + I_2 R_3 \quad ; \ \beta \rangle\rangle 1$$

$$V_{BE1} - V_{BE2} = V_T \ln\left(I_1 / I_2\right)$$

$$V_o = V_{BE3} + \left(R_2 / R_3\right) V_T \ln\left(I_1 / I_2\right) \quad ; \ V_{BE3} = V_T \ln\left[I_3 / I_s\right] \ assume\ room\ temp$$

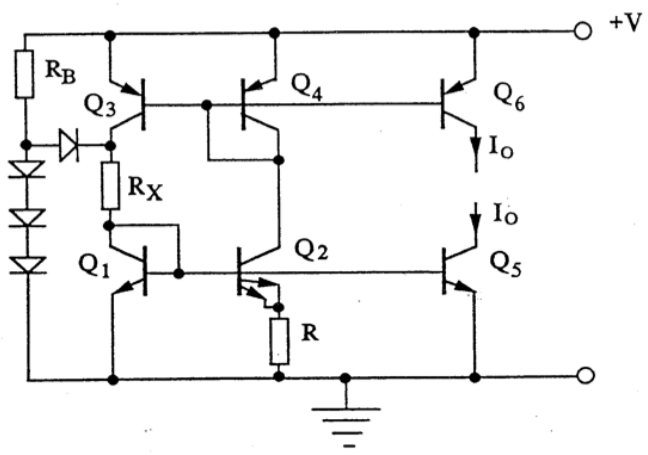
$$For \ \ \partial V_o / \partial T = 0 \Rightarrow \partial V_{BE3} / \partial T = \frac{V_T}{T} \frac{R_2}{R_3} \ln\left[\frac{I_1}{I_2}\right]$$

$$Since \ \partial V_{BE} / \partial T = -2.5\, mV/^{\circ}C \ \ and \ \ \frac{V_T}{T} = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}}$$

$$\Rightarrow \left(\frac{R_2}{R_3}\right) \ln\left(\frac{I_1}{I_2}\right) = 29 \Rightarrow V_o = 1.283$$

[7]

c) Self biased PTAT current source with start up circuit.



Drawing of PTAT [5]
Drawing of Startup Circuit [4]

Answers to question 2:

2. a.

$$Av1 = -gm2/(go2+go4)$$

$$(go2+go4) = I_{D2}(\lambda_N+\lambda_P) = 5\times10^{-6}\times0.05 = 2.5\times10^{-7}\Omega^{-1}$$

$$gm2 = 2\sqrt{\beta_2 I_D}$$

$$\beta_2 = \frac{K_N}{2}\left(\frac{W}{L}\right)_2 = 7.5\times10^{-5}A/V$$

$$gm2 = 3.87\times10^{-5}S$$

$$Av1 = -154.9$$

$$Av2 = -gm6/(go7+go6)$$

$$(go7+go6) = I_{D6}(\lambda_N+\lambda_P) = 20\times10^{-6}\times0.05 = 10\times10^{-7}\Omega^{-1}$$

$$gm6 = 2\sqrt{\beta_6 I_{D6}}$$

$$\beta_6 = \frac{K_N}{2}\left(\frac{W}{L}\right)_6 = 1.6\times10^{-4}A/V$$

$$gm6 = 1.13\times10^{-4}S$$

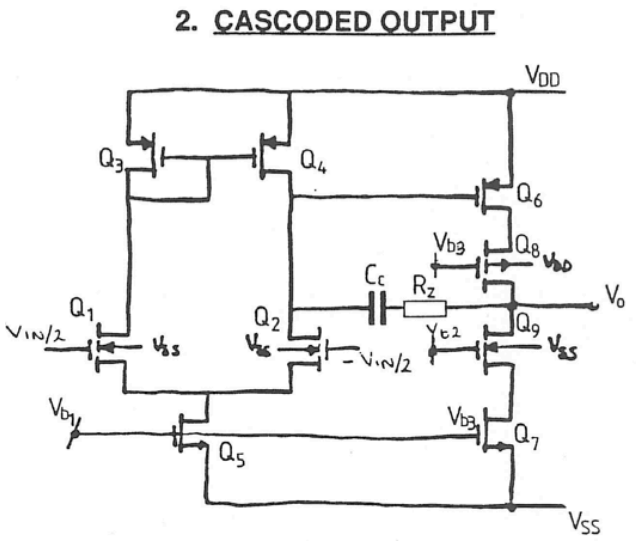
$$Av2 = -113$$

$$A_{total} = Av1\times Av2 = 17503$$

$$G.Bp = gm2/2\pi C_c = 12.32MHz$$

[8]

b.



[4]

Advantages:

Cascoding output stage increases the output impedance by:

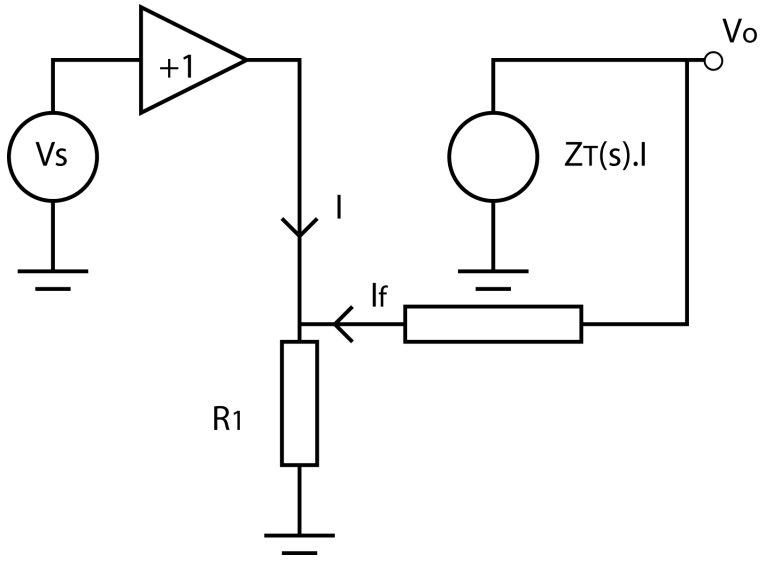
Go=[go6(go8/gm8) +go7(go9/gm9)]

Since Gain of output stage=gm6/[Go]=A_{V2}

Therefore we increase the output impedance and the gain. [2]

Disadvantages:
We have reduced output voltage headroom due to the 4 transistors which need to be in saturation.
Additional bias circuitry is needed to bias the cascade transistors. [2]

c) Current feedback amplifier



$$Z_T(s) = {Z_{T0}}\Big/{\left(1 + jf/f_p\right)}$$

Where fp is the dominant pole
From model

$$I_f = {\left(V_o - V_s\right)}\Big/{R_f}$$

$$I_1 = {V_s}\Big/{R_1}$$

$$V_o = Z_T I = Z_T\left(I_1 - I_f\right)$$

$$V_o/V_s = \left(1 + R_f/R_1\right)\left[\frac{Z_T(s)}{R_f + Z_T(s)}\right] = \left(1 + R_f/R_1\right)\left[\frac{Z_{T0}}{R_f + Z_{T0}}\right]\left(\frac{1}{1 + j f \Big/ f_p\left(\frac{Z_{T0} + R_f}{R_f}\right)}\right)$$

Assuming ZT0>>Rf

$$\left(V_o/V_s\right)_{j\omega} = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{1}{1 + jf\Big/{GB}/{R_f}}\right) \quad ; GB = f_p Z_{T0}$$

$$f_{p-closed} = {GB}/{R_f}$$

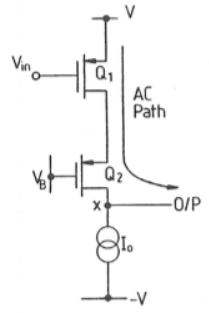
$$A_{closed} = \left(1 + \frac{R_f}{R_1}\right)$$

Therefore the bandwidth is defined by Rf and gain by R1

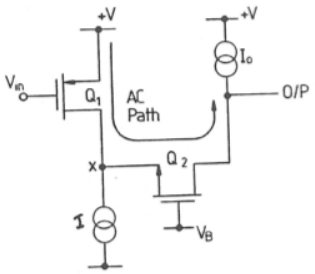
Answers to question 3:

3.a

Concept



Conventional 'stacking' cascode



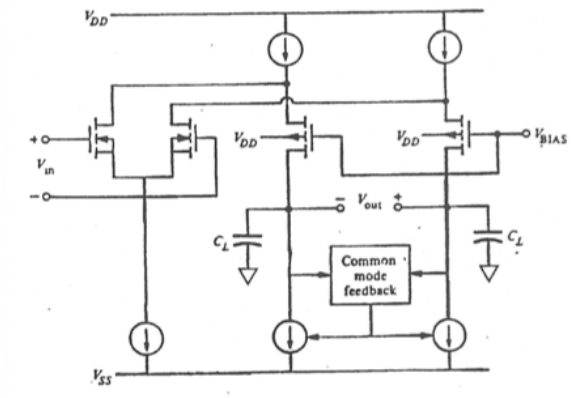
Folded cascode

In the folded cascode we are unstacking the conventional cascode and spreading it out. The AC current path is folded and this allow a reduction in power supply. The conventional cascode requires a 2-stage architecture and since the impedance at (x) is high requires internal compensation. The folded cascode can be used as a single stage architecture, node x is low impedance so the only high impedance node will be at the output.

Gain of the folded cascode Av=gm1/Go.

[4]

b.



[6]

c)

Circuit is a differential continuous time integrator with a balanced double differential linear active transresistor.

$$I_1 = 2\beta[(V_g - V_x - V_T)(V_{in}/2 - V_x) - 1/2(V_{in}/2 - V_x)^2]$$

$$I_2 = 2\beta[(V_g - V_x - V_T)(-V_{in}/2 - V_x) - 1/2(-V_{in}/2 - V_x)^2]$$

$$R = 2V_{in}/(I_1 - I_2)$$

$$R = \frac{1}{\beta(V_g - V_T)}$$

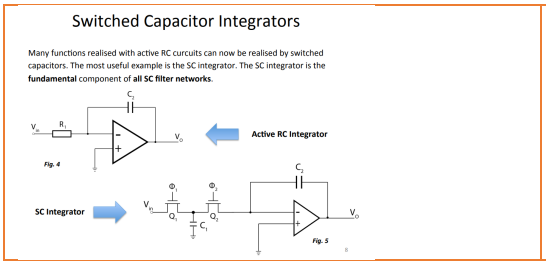
[5]

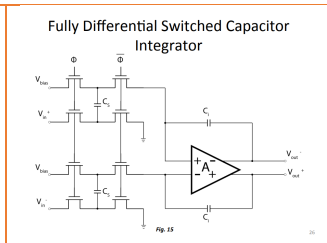
Time constant τ=RC=10ms
R=1/20ux5(1-0.5)/2=40 KΩ
C=τ/R=10m/40K=2.5x10^{-7} f

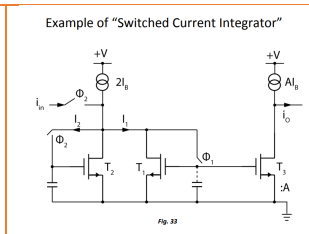
[5]

Answers to question 4:

(a) Sketch circuits for three integrators – A fully differential continuous-time integrator, a switched-capacitor integrator and a switched-current integrator.

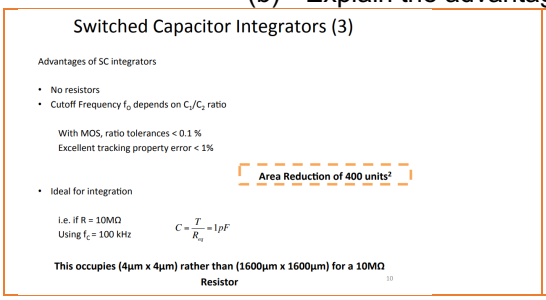


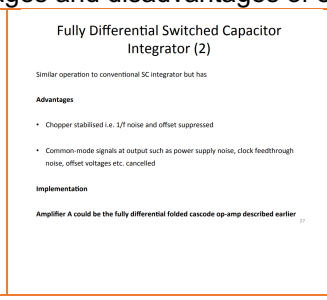


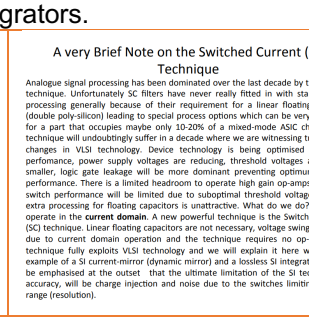


[4]

(b) Explain the advantages and disadvantages of each of the integrators.







[6]

(c) Derive the integrator transfer function of each of the integrators. For both the switched-capacitor and switched-current integrators, what limits the maximum operating frequency?

[10]

Switched Capacitors

$$\frac{V_O}{V_{in}} = \frac{1}{j\omega C_2 R_{eq}} = \frac{f_C C_1}{j\omega C_2} = \frac{1}{jf/f_O} \quad \text{Where} \quad f_O = \left[\frac{1}{2\pi}\right]\frac{C_1}{C_2 . f_C}$$

Time constant $\tau = \frac{1}{f_C}\frac{C_2}{C_1}$

Differential Integrator (3)

$$V_O = \frac{1}{jf/f_O}(V_2 - V_1) \qquad f_O = \frac{1}{2\pi}\frac{C_1}{C_2} f_c$$

OPERATION

Φ_1 $\quad Q = C_1(V_1 - V_2)$

$$I_{av} = f_C C_1 (V_1 - V_2)$$

Φ_2 $\quad I_{av\{Op-amp\}} = -f_C C_1 (V_1 - V_2)$

$$V_O = \frac{1}{j\omega C_2} f_C C_1 [V_1 - V_2]$$

Example of "Switched Current Integrator" (3)

Taking Z transform gives:

$$H(Z) = \left(\frac{i_O}{i_{in}}\right) Z = \frac{AZ^{-1}}{1 - Z^{-1}} \qquad \therefore H(Z) = \frac{A}{Z^1 - 1}$$

$Z^1 \approx (1 + j\omega T...)$ **Assuming ωT << 1, where T is clock period**

$$\therefore H(Z) = \frac{1}{j\omega\frac{T}{A}}$$ **LOSSLESS INTEGRATOR**

$$\tau = \frac{T}{A} = \frac{1}{f_C A}$$ **Time Constant**

Book Reference
"Analogue IC Design: The Current-mode Approach" (Chapter 2), Eds C. Toumazou, F.J. Lidgey and D.G. Haigh

Answer to question 5

a)Derivation from notes

Termed "lossy" since only acts as an integrator at high frequency f > f_o

Verify above transfer function?

SOLUTION

$$\Phi_1 \qquad I_{av} = \frac{C_1 V_{IN}}{T}$$

$$\Phi_2 \qquad \frac{-C_1 V_{IN}}{T} = V_O\left[j\omega C_2 + \frac{C_3}{T}\right]$$

Rearranging yields

$$\frac{-C_1 V_{IN}}{C_3} = V_O\left[\frac{j\omega C_2 T}{C_3} + 1\right]$$

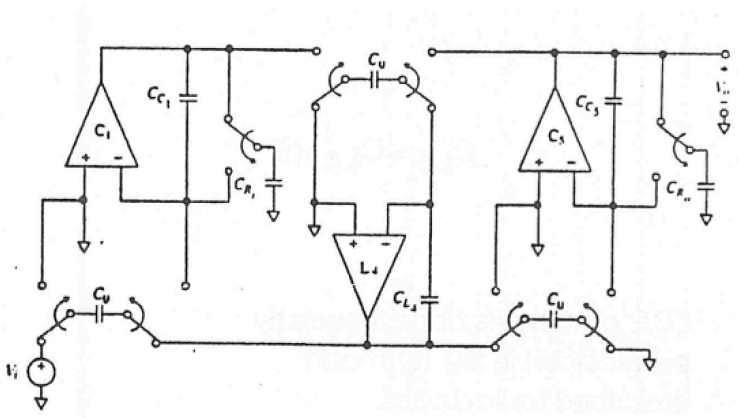
$$\frac{V_O}{V_{IN}} = -\frac{C_1}{C_3}\frac{1}{\left[1 + \frac{j\omega C_2 T}{C_3}\right]}$$

$$= -\frac{C_1}{C_3}\frac{1}{\left[1 + \frac{jf}{f_o}\right]}$$

Where $\quad f_O = \frac{1}{2\pi}\frac{C_3}{C_2} f_C$

[10]

b. (i) 3rd-order Chebyshev low pass switched-capacitor ladder filter.



[5]

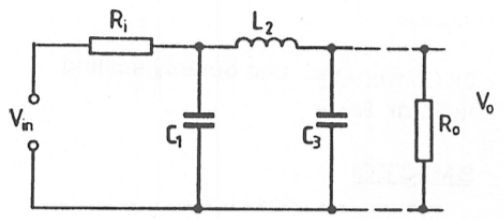
General transformation rules for ladder prototypes:

Inductor: $\frac{f_c L_2}{R_s} = \frac{C_{L2}}{C_u}$

Capacitor: $\frac{C_{c3}}{C_u} = f_c R_s C_3$

Resistor Rs=dummy scalar.

The circuit is equivalent to an RLC prototype



[5]

ii. For switched capacitor equivalent:

C_{c1}=Cc3=5.08 pF

C_{L2}=3.49 pF.

C_u=1pF

Assume scaling Rs=Ri=Ro=1Ω

Therefore

L2= CL2/fc= $3.49/100x10^3=3.49x10^{-5}$ H

Normalised 1rad/sec we multiply by $2\pi f_0$

L2=$3.49 x10^{-5}$ x 2π x 5 KHz =1.096 H

C1=C3=Cc3/fc = $5.08/100x10^3$ =$5.08x10^{-5}$ f

Normalised value C1=C3=1.596

Answer to question 6

(a) Sketch the circuit of a VGS multiplier with equations describing its operation.

V_{GS} Multiplier

Stolen from the V_{BE} multiplier

Circuit Analysis

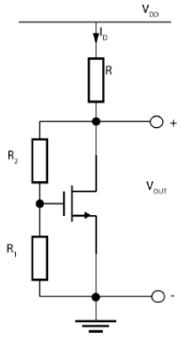


Fig. 16

$$V_{OUT} = V_{GS}\left[1+\frac{R_2}{R_1}\right]$$

OR

$$V_{OUT} = \left[1+\frac{R_2}{R_1}\right]\left(V_T + \sqrt{\frac{V_{DD}}{\beta R}}\right)$$

Note: All R's can be replaced with active resistors

25

[6]

(b) From notes:

Improved Biasing for Maximum Output Swing (2)

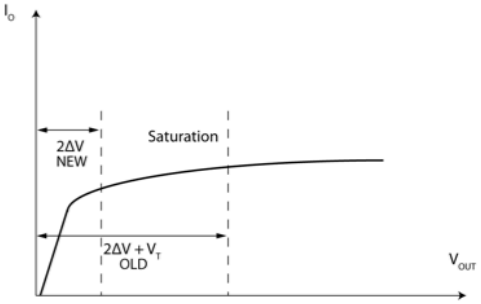
Assuming Equal lengths and $I_O = I_{IN}$ Remember: β = KW/2L

$$I_D = \beta(V_{GS} - V_T)^2$$

$$\beta_1 = \beta_2 = \beta_4 = \beta_5 = \beta_6 = \beta$$

$$\beta_3 = \left[\frac{\beta}{4}\right]$$

Since equal I_D, then for Q_3, $V_{DSsat} = 2(V_{GS} - V_T)$



EXAMPLE

Let $(V_{GS} - V_T) = \Delta V$ and prove to yourselves that

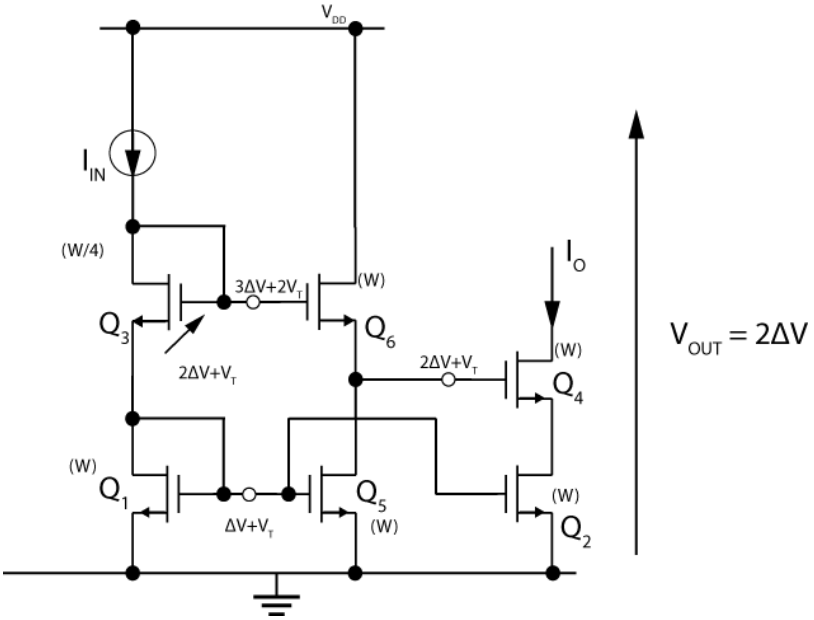
$$V_{OUT(\min)} = 2\Delta V$$

Fig. 32

HINT: *remember that* $V_{DS} = V_{GS} - V_T$ *or* $V_D = V_G - V_T$

52

Annotation of VGS' of each transistor:



(c) Sketch a suitable two transistor and four transistor potential divider voltage reference circuit and show that when VDD=+5V, VSS=-5V and ID=10uA, for an output voltage Vout=0, the four transistor circuit consumes less area than the two transistor one. You may assume K_P=8μA/V^2, K_N=17μA/V^2, V_{TN}=V_{TP}=1V.

[5]

c.b The two and four transistor potential dividers depicted in figure below.

$$K_n = 17\mu A/V^2; K_p = 8\mu A/V^2;\ V_{Tn} = V_{Tp} = 1V; V_{DD} = +5V; V_{SS} = -5V; V_{Out} = 0$$

For the two transistor potential divider

$$I_{Qp} = I_{Qn} = \tfrac{1}{2} K_n \frac{W_{Qn}}{L_{Qn}} \left(V_{out} - V_{ss} - V_{Tn}\right)^2 = \tfrac{1}{2} K_p \frac{W_{Qp}}{L_{Qp}} \left(V_{out} - V_{dd} - V_{Tp}\right)^2$$

$$\Rightarrow \frac{W_{Qn}}{L_{Qn}} = \frac{10}{136} = 0.073; \; \frac{W_{Qp}}{L_{Qp}} = \frac{10}{64} = 0.156$$

For the four transistor potential divider

$$I_{Qp1} = I_{Qn1} = I_{Qp2} = I_{Qn2} = \tfrac{1}{2} K_n \frac{W_{Qn1}}{L_{Qn1}} \left(V_{GSn1} - V_{Tn}\right)^2 = \tfrac{1}{2} K_n \frac{W_{Qn2}}{L_{Qn2}} \left(V_{GSn2} - V_{Tn}\right)^2$$

$$= \tfrac{1}{2} K_p \frac{W_{Qp1}}{L_{Qp1}} \left(V_{GSp1} - V_{Tp}\right)^2 = \tfrac{1}{2} K_p \frac{W_{Qp2}}{L_{Qp2}} \left(V_{GSp2} - V_{Tp}\right)^2$$

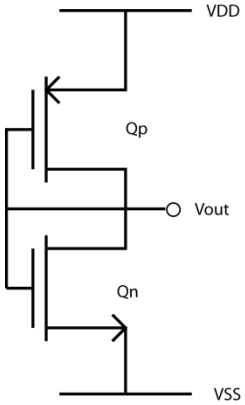
$$V_{GSn1} = V_{GSn2} = \frac{V_{Out} - V_{SS}}{2} = 2.5V$$

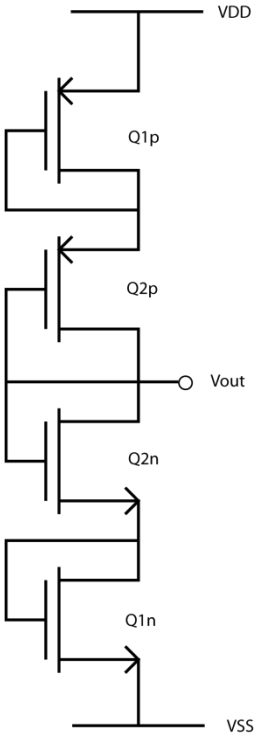
$$V_{GSp1} = V_{GSp2} = \frac{V_{Out} - V_{DD}}{2} = 2.5V$$

$$\frac{W_{n1}}{L_{n1}} = \frac{W_{n2}}{L_{n2}} = \frac{10}{19.125} = 0.523$$

$$\frac{W_{p1}}{L_{p1}} = \frac{W_{p2}}{L_{p2}} = \frac{10}{10} = 1$$

The aspect ratio of the transistors in the four-transistor potential divider is much less than the two-transistor circuit, therefore, its consuming area will be less. For example, assuming common width of 10 units, for the two transistor configuration the active area becomes 10*(64+136)=2000 units while for the four transistor it becomes 10*2*(10+19.125)=582.5 units.





[Total 7 points]